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Control System I Laboratory

Final Project Report

Section: C2 Group: 05

A Multi-Sensor Controlled 6-DOF Robotic Arm for Human Motion Replication

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1 Abstract

This project focuses on developing a gesture-based 6-DOF robotic arm that can follow human arm movements in real time. Instead of using traditional input devices like joysticks, we used multiple motion sensors placed on the human arm to capture natural movements. These signals are processed and translated into corresponding motor actions.

The system combines accelerometer, gyroscope, magnetometer, and Hall-effect sensor data to determine different joint angles such as shoulder, elbow, wrist, and base rotation. An Arduino-based controller processes these inputs and generates PWM signals to drive the servo motors.

One of the key aspects of this project is achieving smooth and stable motion using filtering techniques and proper angle mapping. The final system is able to perform coordinated movements like lifting, rotating, and gripping objects.

Overall, the project demonstrates a simple yet effective approach to human motion replication, which can be extended to applications like assistive robotics, remote manipulation, and training systems.

2 Introduction

Controlling machines using natural human motion is always more intuitive than using switches or programmed commands. With that idea in mind, this project attempts to bridge the gap between human movement and robotic control.

We developed a robotic arm that follows the motion of a human arm by collecting real-time data from sensors mounted on different joints. When the user moves their hand, the robotic arm tries to replicate the same motion almost instantly. This makes the interaction feel direct and responsive.

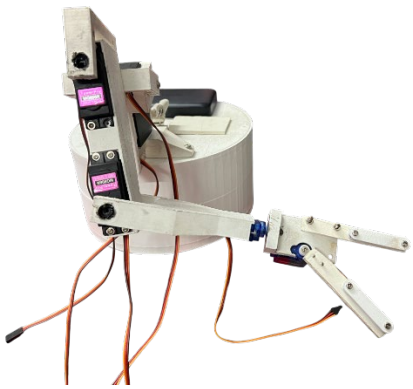

Fig 2.1: Developed Robotic Arm System

The main challenge was not just reading sensor data, but converting that raw data into meaningful angles and stable motor control signals. Since multiple joints are involved, small errors or noise in data can lead to noticeable instability. To overcome this, filtering techniques and proper reference adjustments were used.

Compared to conventional robotic systems, this approach is more engaging and user-friendly. It does not require prior training or complex programming from the user side.

This kind of system can be useful in situations where direct human presence is risky, such as handling hazardous materials, or in assistive technologies where motion needs to be replicated remotely.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

This project focuses on designing a **6-DOF robotic arm** capable of mimicking basic human hand movements using sensor-based input.

Project Scope

- **Robotic Arm Design:**
  Design and construction of a lightweight, modular 3D-printed 6-DOF robotic arm capable of replicating basic human arm movements such as shoulder, elbow, yaw, wrist, and gripper motion.
- **Motion Sensing:**
  Acquisition of human arm motion using IMU sensors and calibration of sensor data to represent joint positions accurately.
- **Data Processing and Mapping:**
  Conversion of sensor readings into joint angles and mapping them to servo PWM signals. Basic filtering and motion smoothing techniques were implemented to improve movement accuracy.
- **Microcontroller Interface and Programming:**
  Communication between sensors, the Arduino Uno microcontroller, and actuators is managed through embedded firmware developed using an IDE for real-time data processing and servo control.
- **Real-Time Operation:**
  The system performs real-time mapping of human motion to robotic motion with minimal delay.

Limitations

- **No Feedback System:**
  The robotic arm operates as an open-loop system without position feedback, which may lead to error accumulation and reduced accuracy.
- **No Wireless Control:**
  The system uses wired connections only, limiting mobility and application range.
- **Sensor Limitations:**
  Magnetometer readings are sensitive to magnetic interference, and the Hall sensor requires close proximity to detect motion accurately.
- **Practical Limitations:**
  The system is designed for demonstration purposes and is not intended for heavy-duty or continuous industrial operation.

3.1.2 Literature Review

Robotic arms and hand-mimicking systems have been widely explored in both research and educational projects. Many existing designs rely on flex sensors attached to fingers that change their resistance values according to the bending and thus control a certain DOF[1]. However, these designs do not completely replicate the anatomically correct joint parts of a human hand.

Previous works also emphasise the use of IMUs, or potentiometers to capture human motion, combined with open-loop or closed-loop control to drive servo motors. These systems focus primarily on demonstrating motion capture and actuation rather than precise replication of joint kinematics.

In contrast, our project aims to physically mimic the human hand's joint structure in the 3D-printed robotic arm. The goal is to mount the sensor on specific joints of the user's hand, and the robot exactly follows what the user is trying to do. To achieve this, we incorporated two new kinds of sensors: the Magnetometer and Hall Sensor, which are not widely used in existing models. While the control strategy and sensors used are based on established methods, the physical replication of joint motion gives our system a closer resemblance to actual human hand movement compared to existing approaches.

3.1.3 Formulation of Problem

The main purpose of the project is to design a 6-DOF robotic arm that replicates human arm and hand movements.

Objectives:

- Design a 6-DOF robotic arm that replicates human arm and hand movements.
- Process sensor data and calculate joint angles in real time.
- A basic understanding of how sensors, data processing, and actuators are integrated to replicate human movements in practical applications such as

prosthetics, teleoperation, and robotics.

Constraints:

- Open-Loop Control: The system does not have feedback from the robotic joints, making it susceptible to errors and deviations in motion.
- Mechanical Limitations: Joint ranges, servo torque, and link lengths restrict full replication of human arm and hand movements.
- Sensor Limitations:
    - IMUs and magnetometers are prone to noise, drift, and interference[2].
    - Hall sensors require close proximity to magnets for accurate readings, limiting the range.
- Wiring Constraints: Multiple wires restrict arm movement and flexibility.
- Power Limitations: Motors require sufficient current; battery capacity and voltage regulation must be managed.
- No Wireless Control: The system operates through direct wired connections, limiting remote operation.
- Calibration Requirements: Accurate motion replication depends on correct sensor alignment, angle offsets, and mapping, which adds complexity.

3.1.4 Analysis

The analysis phase involved evaluating system requirements, selecting appropriate components, and determining the feasibility of implementing the robotic arm.

Functional Analysis

The system performs the following functions:

- Detect hand movement using sensors
- Process sensor data using a microcontroller
- Convert sensor readings into control signals
- Drive servo motors to move the robotic arm
- Replicate human hand motion in real time

These functions ensure coordinated system operation.

Technical Analysis

Key technical considerations include:

- **Sensor Performance:** Sensors must provide accurate and stable motion measurements.
- **Motor Control:** Servo motors must supply sufficient torque and speed for smooth joint movement.
- **Power Supply:** The system must provide stable voltage and current to all components.

- **Mechanical Structure:** The structure must be strong yet lightweight for efficient operation.

Feasibility Analysis

The project was considered feasible due to the availability of low-cost components, simple control algorithms, manageable power requirements, and compatibility between electronic and mechanical subsystems.

3.2 Design Method (PO(a))

3.2.1 Workflow :

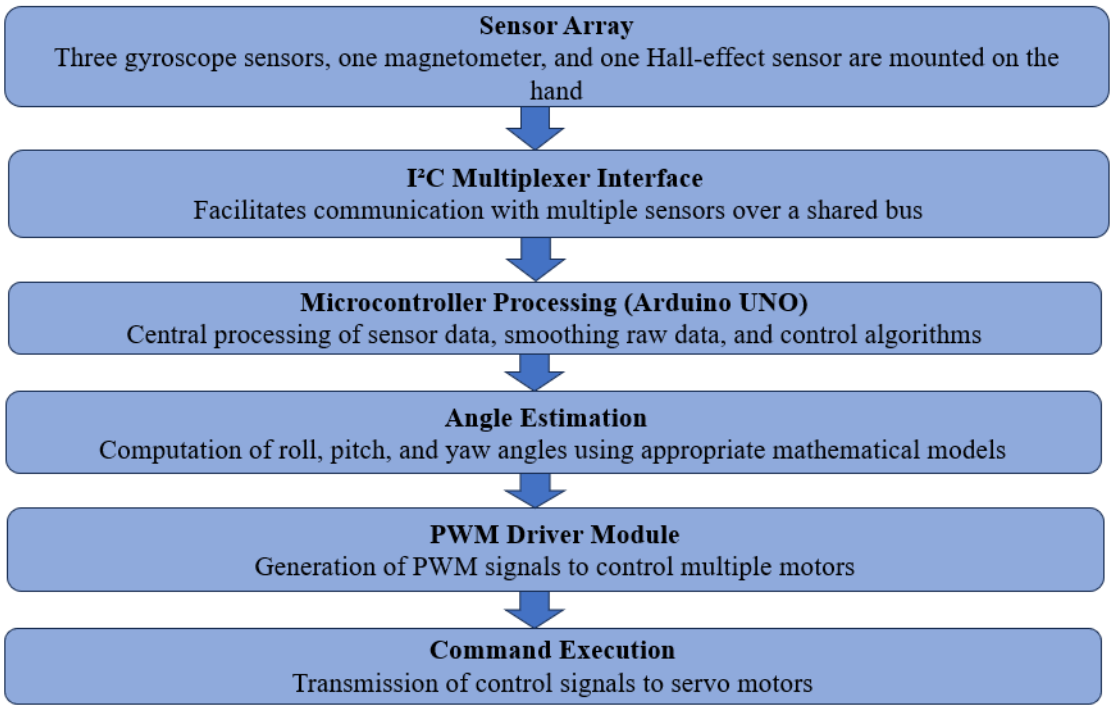

Fig 3.1: Workflow

3.2.2 Problem Handling Methods:

i) Choosing an appropriate power source for powering 6 servo motors:
Running 6 servo motors simultaneously draws a lot of current. Using a normal 5V battery would not be sufficient for our project. So we used a 12 Volt LiPo battery along with a Buck converter, which is able to handle large currents and then stepped it down to 5 Volts and used it as the supply voltage for the motors.

ii) Using an I2C MUX for taking raw data from multiple Gyro sensors at once :
Each MPU6050 gyroscope module shares the same default I2C address[3], which prevents multiple sensors from being connected directly to the Arduino UNO on the same bus. To acquire data from multiple sensors simultaneously, an I2C multiplexer (MUX) is utilised.

The MUX enables selective communication by switching between channels[4], allowing the microcontroller to access each sensor individually despite having identical addresses. This facilitates efficient and sequential data acquisition from all connected sensors.

iii) A PWM motor driver to run motors without stressing the microcontroller:
A PWM driver board was used to control multiple servo motors efficiently using a single microcontroller. It generates stable PWM signals independently and allows multiple motors to be controlled via I2C communication[5], reducing the computational load on the microcontroller and simplifying the overall system design.

iv) Smoothing out the raw data and handling jittering:
The raw data from the sensors can be pretty unreliable. To make the motion smoother, we used the Moving Average method so that the data becomes more stable. Also, to ignore small fluctuations, a deadband was used, so any angle fluctuations below that threshold are completely ignored.

v)Adjusting the zero positioning of the motors and the correct orientation of the sensors :
The servo motors used in the system operate within a range of **0° to 180°**. To ensure we are getting the correct range of motion, we had to assemble the arm parts by correctly understanding where the zero position of the motors should be before joining them. Also, we had to figure out in which orientation our sensors should be, which formulas to use for which joint and if we need to add any offsets to the servo angles.

3.2.3 Angle Measurement and Algorithm Analysis:

A. Shoulder Joint:

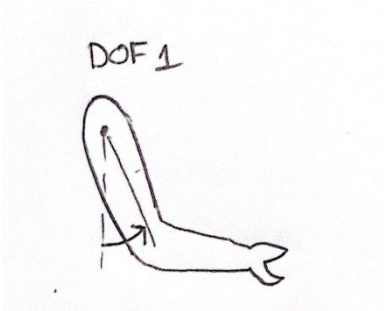

Fig 3.2a: Shoulder pitch

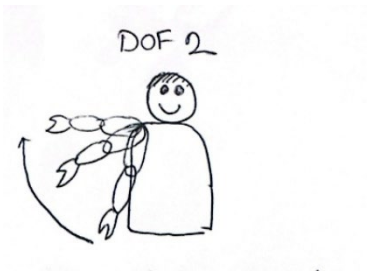

Fig 3.2b: Sideways elevation

A single gyroscope is mounted on the upper arm to measure pitch motion and lateral (side-to-side)
elevation of the arm. For this particular orientation of the Gyro on the hand (the first figure), we figured out which components of the accelerometer we should consider for angle measurement. The formulas for pitch are

$$Shoulder\ Pitch = tan^{-1}\left(\frac{a_x}{\sqrt{a_y^2+a_z^2}}\right)$$

$$Sideways\ Elevation\ Angle = tan^{-1}\left(\frac{a_z}{\sqrt{a_x^2+a_z^2}}\right)$$

B. Elbow Joint:

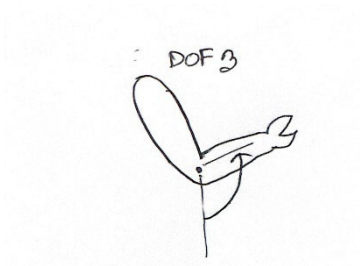

Fig 3.3: Elbow Pitch

One gyroscope is mounted right after the elbow joint in the lower arm. Here, we calculated the elbow pitch, but since the shoulder angle is measured from the vertical reference line and moving the shoulder joint also moves the elbow, we subtracted shoulder pitch from elbow pitch to find the actual rotation angle of the servo motor at the elbow joint. The formulas used here are,

$$Elbow\ Pitch = tan^{-1}\left(\frac{a_y}{\sqrt{a_x^2+a_z^2}}\right)$$

$$Elbow\ Angle = \ Elbow\ Pitch\ -\ Shoulder\ Pitch$$

Here is a combined flowchart for shoulder and elbow angle measurement:

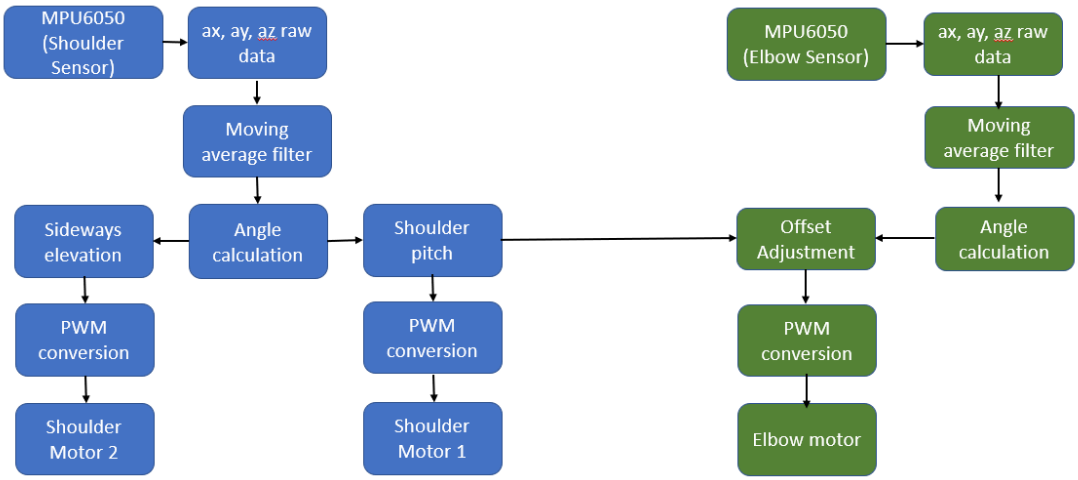

Fig 3.4: Algorithm Flowchart for Shoulder and Elbow Angle measurement

C. Base Control:

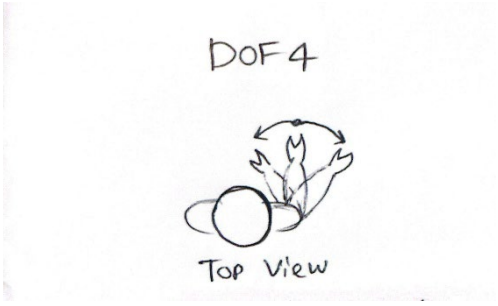

Fig 3.5: Yaw Movement

A magnetometer is used to control the base rotation of the robotic arm by mapping the hand's horizontal yaw. The initial Earth's magnetic field orientation is set as the reference (90° servo position), and the current azimuth angle is compared to this reference to determine relative left–right orientation. To ensure the roll and pitch of the hand do not interfere with the horizontal azimuthal angle, we used a compensated formula for azimuth-

$X_h = Mx \cos\theta + Mz \sin\theta$
$Y_h = Mx \sin\varphi \sin\theta + My \cos\varphi - Mz \sin\varphi \cos\theta$
Base Angle $= \tan^{-1} (Y_h / X_h)$
Here,
X_h and Y_h correspond to the horizontal components of the magnetometer data
θ = Elbow Pitch
φ = Roll angle;

The Algorithm flowchart for the magnetometer is given below

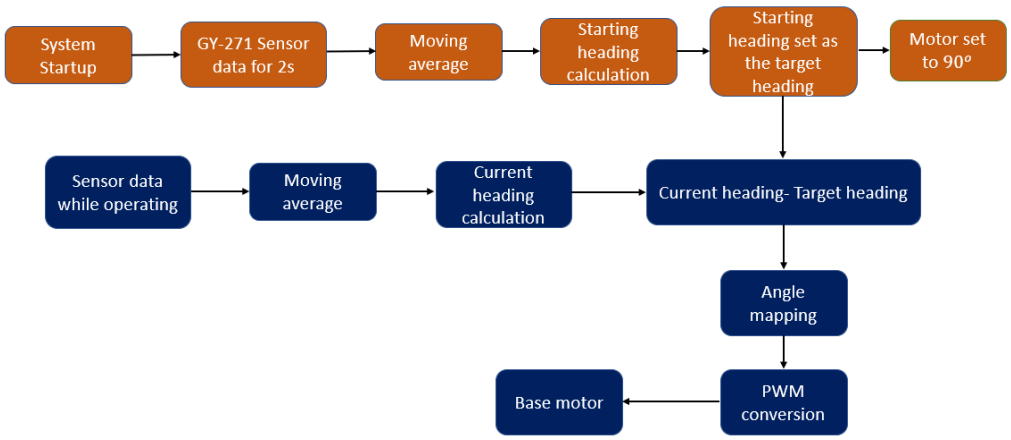

Fig 3.6: Algorithm flowchart for magnetometer

D. Wrist Control:

Fig 3.7: Wrist roll

The wrist is controlled by calculating the Roll angle from the Gyro sensor mounted on the wrist. The mathematical formula here used is-

$$Roll = tan^{-1}\left(\frac{a_y}{a_z}\right)$$

The algorithm flowchart for wrist movement is as follows

Fig 3.8: Algorithm Flowchart for Wrist

E. Gripper Control:

Fig 3.9: Gripper movement

The gripper opening is controlled by measuring magnetic distance using the Hall sensor and mapping that value to a servo angle. The analogue data obtained at two extreme – i) Magnet farthest and ii)Magnet closest is taken as the two limits of the angle to be mapped;

The algorithm flowchart for gripper control is as follows

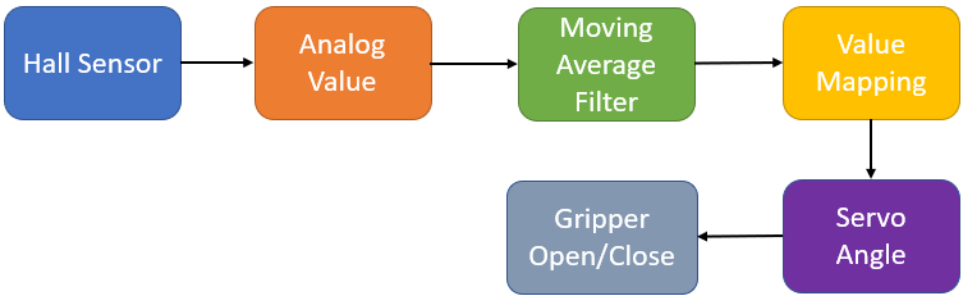

Fig 3.10: Algorithm flowchart for Gripper

3.2 Circuit Diagram

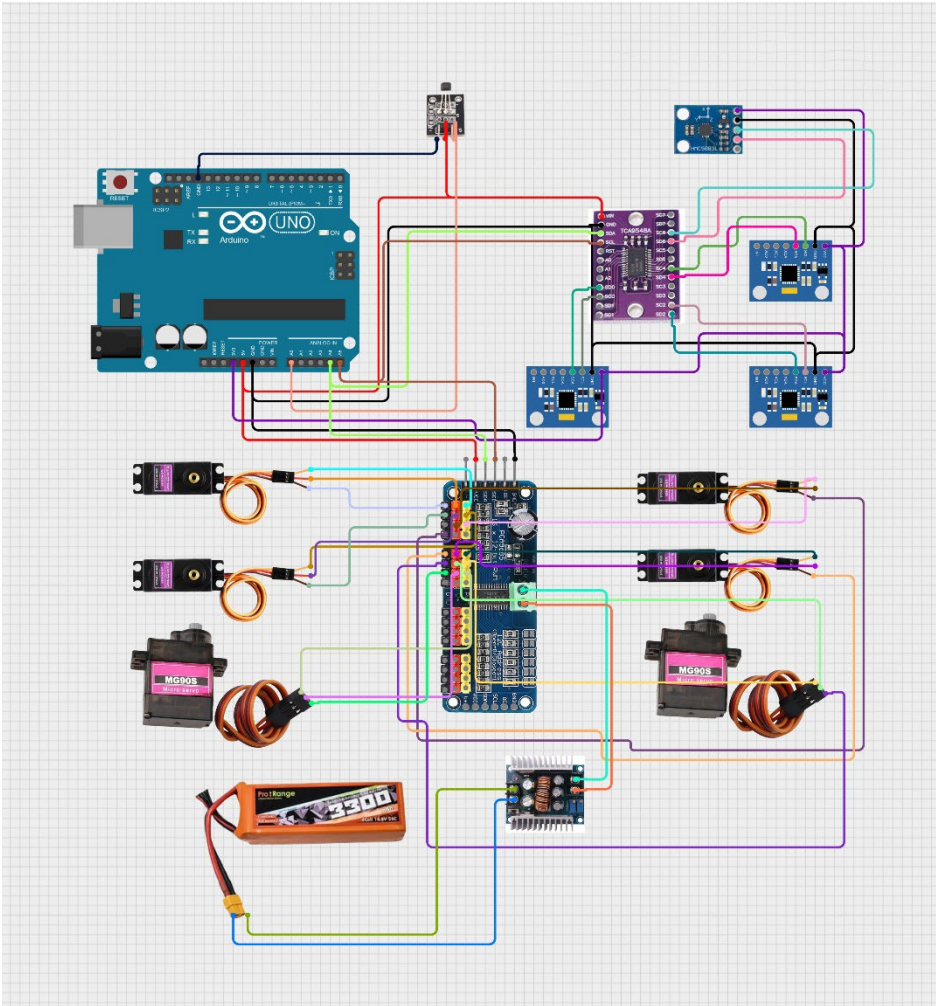

Fig 3.11: Circuit Diagram

3.3 Circuit Design

After building the circuit on the breadboard, instead of designing a PCB, we transferred it to a Veroboard and soldered the connections using connecting wires and used header sockets for holding the pin heads or the jumper wires. This significantly reduced the complexity of the circuitry.

3.4 Full Source Code of Firmware

```
#include <Wire.h>
#include <MPU6050.h>
#include <QMC5883LCompass.h>
#include <Adafruit_PWMServoDriver.h>
#include <math.h>
//////////////////////////////////////////////////////////
// OBJECTS
//////////////////////////////////////////////////////////
MPU6050 mpuShoulder;
MPU6050 mpuElbow;
MPU6050 mpuWrist;
QMC5883LCompass compass;
Adafruit_PWMServoDriver pwm =
    Adafruit_PWMServoDriver();
//////////////////////////////////////////////////////////
// MUX SETTINGS
//////////////////////////////////////////////////////////
#define TCAADDR 0x70
void tselect(uint8_t i)
{
  if (i > 7) return;
  Wire.beginTransmission(TCAADDR);
  Wire.write(1 << i);
  Wire.endTransmission();
delayMicroseconds(200);
}
```

Find the rest of the code at GitHub : https://github.com/intmushfiq/6-DOF-Robotic-Arm/blob/main/final_leap_of_faith.ino

4 Implementation

4.1 Description

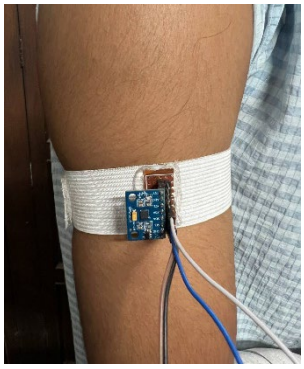

Fig 4.1a: Sensor Placed on Shoulder

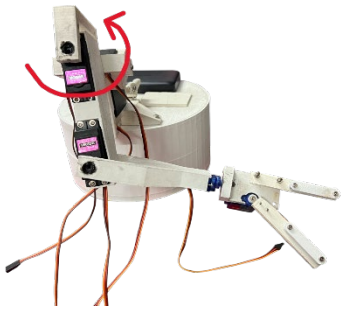

Fig 4.1b: Shoulder Joint

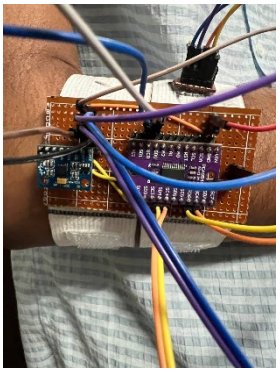

Fig 4.1c: Sensor Placed on Elbow

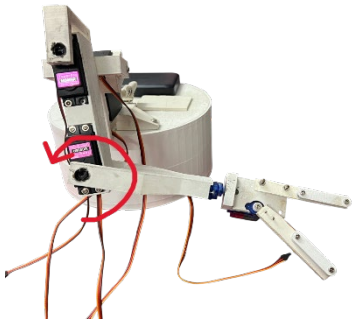

Fig 4.1d: Elbow Joint

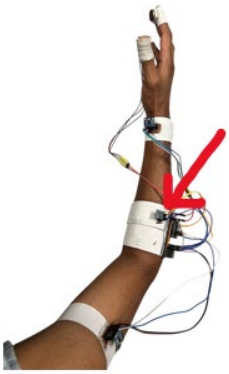

Fig 4.1e: Magnetometer placement

Fig 4.1f: Base Joint

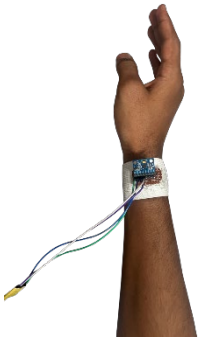

Fig 4.1g: Sensor Placed on Wrist

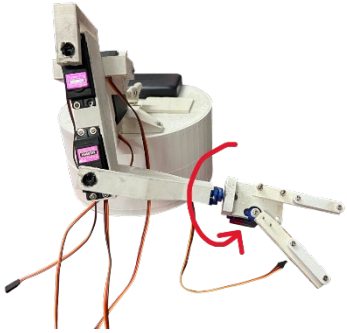

Fig 4.1h: Wrist Joint

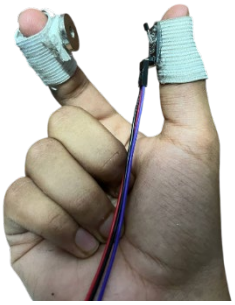

Fig 4.1i: Hall Sensor Placement

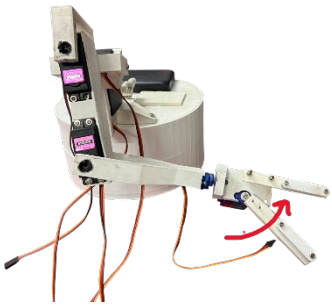

Fig 4.1j: Gripper Joint

5 Design Analysis and Evaluation

5.1 Novelty

The proposed system features a multi-sensor robotic arm that mimics human hand movements by integrating motion sensing, orientation tracking, and grip detection to operate in real time. Unlike traditional robotic arms that depend on manual joysticks or programmed commands, this design utilizes natural hand gestures captured via wearable sensors to control the robotic joints.

The novelty of the system lies in the following aspects:

- Multi-sensor fusion approach:
  The system simultaneously utilizes:
  - MPU6050 sensors for shoulder, elbow, and wrist motion detection
  - GY-271 compass sensor for left–right rotational movement
  - S49E Hall effect sensor for gripper control

  These sensors collectively capture different degrees of freedom of the human hand and translate them into robotic motion.

- Real-time gesture replication:
  The robotic arm mimics human motion instantly by processing sensor data and mapping it to servo motor angles.
- Cost-effective implementation:
  The system uses low-cost microcontrollers and commercially available sensors while maintaining acceptable precision and responsiveness.
- Modular architecture:
  The design allows easy expansion, such as adding more sensors or increasing degrees of freedom.

The robotic arm is designed to replicate shoulder, elbow, wrist, base rotation, and gripper movements using multiple servo motors controlled through a PWM driver. This combination of gesture-based control, sensor fusion, and low-cost hardware implementation demonstrates the innovative nature of the project.

5.2 Design Considerations (PO(c))

The design of the robotic arm considered factors related to public safety, environmental impact, and societal usability to ensure safe, reliable, and socially acceptable operation.

5.2.1 Considerations to public health and safety

Public safety was a key consideration during system design.

- **Electrical Safety:** Low-voltage components were operated within safe limits using voltage regulators, proper insulation, and safe battery handling procedures.

- **Mechanical Safety:** Safety was ensured by limiting servo rotation angles, using lightweight 3D-printed parts, and securely fastening mechanical joints.

5.2.2 Considerations to environment

Environmental sustainability was considered during component selection and system design.

- **Energy Efficiency:** Low-power components were used to reduce energy consumption and extend battery life.
- **Material Selection:** 3D-printed parts and reusable electronic components were used to reduce material waste.
- **Electronic Waste Management:** Replaceable components were used to minimize electronic waste.

5.2.3 Considerations to cultural and societal needs

The robotic arm was designed to address societal needs and improve usability.

- **Assistive Technology:** The system can assist disabled individuals, elderly persons, and workers in hazardous environments by enabling remote task execution.
- **Educational Value:** The project supports learning in robotics, embedded systems, and control systems, promoting STEM education.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The objective was to evaluate the accuracy and responsiveness of the robotic arm in replicating human hand movements.

Experimental Setup
The system included an Arduino Uno microcontroller, MPU6050 sensors, GY-271 compass sensor, S49E Hall effect sensor, MG90S and MG996R servo motors, a PCA9548A I2C multiplexer and a PCA9685 PWM driver. Sensor data were processed to control the servo motors in real time.

Test Conditions
At first, all the sensors, modules and servo motors were tested individually. Movements tested included shoulder lifting, elbow bending, wrist rotation, base rotation, and gripper operation. Each movement was repeated multiple times to ensure consistency.

5.3.2 Data Collection

Sensor data were collected through the Arduino microcontroller using serial communication.

Measured Parameters:

- Accelerometer's raw axis values
- Analog value of the hall sensor when a magnet was at the nearest and farthest
- Servo position angles
- Movement accuracy

5.3.3 Results and Analysis

The robotic arm replicated human movements with acceptable accuracy and stability. Relatively smooth motion was achieved using filtering techniques.

5.3.4 Interpretation and Conclusions on Data

The results indicate that the system can accurately detect human motion, process sensor data effectively, and control servo motors reliably, making it suitable for gesture-based robotic applications.

5.4 Limitations of Tools (PO(e))

Sensors Limitations

- The hall sensor works in a close range, hence the gripper motion is only replicated when the fingers are in close proximity. Also, temperature variation can drift measurements. [6]

- The accelerometers' measurement drift over time. The sensors are also very sensitive to vibration.

- The magnetometers' measurements are highly affected by nearby magnetic interferences. [7]

Servo Motors Limitation

- The motors consume high current, requiring a high current-rated buck converter, which is extremely delicate.[8]

- The micro servo used for the wrist and gripper has limited torque capacity

System-Level Limitations

- Limited payload capacity
- Sensor noise and calibration errors
- Mechanical wear over time
- Battery discharge limitations

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

From a broader perspective, this project reflects how robotics can gradually become part of everyday life. A system like this can be used in education to make learning more interactive and practical. Students can actually see how their movement translates into machine action, which builds a better understanding.

In terms of societal impact, such robotic arms can be helpful for people with limited mobility. With further development, this idea can be extended to assistive devices where human motion is replicated mechanically.

Culturally, the project promotes innovation using low-cost components, which is important in a developing country context. It shows that meaningful engineering solutions can be built without expensive infrastructure.

5.5.2 Assessment of Health and Safety Issues

Safety was an important consideration during the project. Since the system involves moving mechanical parts, there is always a risk of sudden or unintended motion. To reduce this, we ensured the controlled speed of servo motors and avoided extreme angle jumps.

At the same time, one positive impact is that such a system can reduce human exposure to dangerous environments. For example, instead of directly handling harmful objects, a robotic arm can be used as an intermediate tool.

5.5.3 Assessment of Legal Issues

The project complies with standard engineering practices and safety guidelines. Legal considerations include safe handling of lithium batteries, proper disposal of electronic waste, and compliance with electrical safety standards. No hazardous or restricted materials were used in the system.

If this system is developed further for commercial use, then safety standards, certifications, and usage regulations would need to be considered.

5.6 Sustainability Evaluation (PO(g))

While designing the system, we tried to keep resource usage reasonable. The components used are mostly low-power devices, which keep overall energy consumption relatively low.

The use of 3D printing helped reduce material wastage, since parts were produced as needed instead of mass fabrication. Also, most of the electronic components can be reused in future projects, which adds to sustainability.

One concern is battery usage, especially LiPo batteries, which need proper handling and disposal. Care must be taken so that these components do not harm the environment after their lifecycle.

In general, the project does not create a significant environmental burden and can be considered moderately sustainable for a prototype-level implementation.

5.7 Ethical Issues (PO(h))

The 3D model of the robotic arm was sourced from an open-source platform, "Electromaker[9]." Also, the circuit diagram was designed using an open-source circuit designing web app named "Cirkit Designer[10]."

Throughout the project, we maintained honesty in both design and reporting. All external resources, including the robotic arm model, were properly referenced. No part of the work was copied without acknowledgment.

Another ethical consideration was safety. We avoided designing anything that could harm users during operation. The system was tested carefully before the demonstration.

We also ensured that the project remains educational and constructive in nature. While robotics can sometimes be misused, our focus was strictly on learning, innovation, and beneficial applications.

Working as a team, we divided tasks fairly and ensured that each member contributed. This also reflects ethical teamwork and responsibility.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Student ID	Name	Contribution
2106177	**Tonmoy Deb**	Hardware troubleshooting, circuit designing, component buying and managing, presentation and report writing
2106181	**Mushfiqur Rahman**	Gripper design modification, gripper code implementation, wrist code implementation, Sensor holder soldering, presentation and report writing
2106182	**Chayan Paul**	PWM driver and I2C Mux operating, hardware troubleshooting, components buying and managing, system debugging and testing
2106183	**Shamin Yeaser**	Mathematical analysis, shoulder and elbow code implementation, base logic adjustments, presentation and report writing
2106184	**Kayes Sami Malitha**	Base code implementation, hardware troubleshooting, arm band for sensor holder managing, system debugging and testing

6.2 Mode of Team Work

- Tasks were divided based on individual strengths in hardware efficiency and code implementation
- Regular meetings were held to review progress

- Code integration and debugging were performed collaboratively
- Team members supported each other during testing and troubleshooting

6.3 Diversity Statement of Team

The team consisted of members with different technical expertise and skills in hardware design, Arduino programming, and data analysis. This diversity helped in approaching the problem from multiple perspectives and contributed to the successful implementation of the robotic arm system. Collaboration among team members allowed effective problem-solving and improved the overall quality of the project.

6.4 Log Book of Project Implementation

Week	Milestone achieved	Individual Role	Team Role	Comments
Week 4	Literature review on robotic arm model	2106177 and 182 reviewed the arm model from various sources	Team discussed the mechanical system architecture	Mechanical design approach finalized
Week 5	accelerometer-based roll-pitch calculation, hall sensor and magnetometer	2106183 reviewed angle calculation, 2106181 reviewed hall sensor application and 2106184 reviewed magnetometer application	Team discussed electrical system architecture	Electrical system approach finalized
Week 6	Mechanical system assembly and troubleshooting	2106182 and 2106177 lead the arm assembly and other members helped	The team discussed problems with the mechanical model	Initial assembly of mechanical model completed
Week 7	Mechanical system finalization	2106183, 2106182 and 2106184 modified the joints of the arm	The team verified the final mechanical system	Mechanical system finalized
Week 8	Servo motors were run using individual sensors	2106181 worked with the hall sensor, 183 with the accelerometer and 184 with the magnetometer	Team verified sensor operation and calibration	Sensor data recorded and calibrated
Week 9	Electrical system troubleshooting	2106181 realized the need of I2C mux, and 2106182 bought and implemented multiple sensors with the mux	Team verified correct operation and helped fix problems with the electrical system	Electrical system ready
Week 10	Electrical system assembly	2106177 designed the circuit, 2106184 made arm bands for the sensor holder, and 2106181 soldered the sensor holder on a Veroboard using header pins	Team verified correct connections	Electrical system implemented

Week 11	Arm movement replicated through logical code fixing	2106183 implemented shoulder and elbow, 184 implemented base, and 181 implemented wrist and gripper code	Team observed correct arm movement	Initial code ready, encountered a problem with the gripper model
Week 12	Gripper model fixing and all code merging	2106181 modified the gripper model and 2106183 and 182 merged the code. 2106177 helped with debugging	Team verified correct operation	Final model ready
Week 13	Final debugging and documentation	All members participated	Team finalized documentation and results.	Project completed

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

A gesture-controlled robotic arm has been developed using low-cost sensors and microcontroller technology to replicate human hand movements in real time. The system uses motion sensors to detect shoulder, elbow, wrist, and hand movements, which are then translated into robotic arm motion using servo motors. This innovative technology can assist people with disabilities, support automation in industries, and improve safety in hazardous environments. The robotic arm is energy-efficient, portable, and easy to operate. By combining affordability with functionality, the project demonstrates how modern robotics can be made accessible for education, healthcare, and everyday practical applications.

7.2 User Manual

System Overview

The robotic arm is a gesture-controlled system designed to mimic human hand movements. Sensors attached to the user's hand detect motion, and the microcontroller processes the data to control servo motors accordingly.

Main Components:

- Arduino Uno microcontroller
- MPU6050 motion sensors
- GY-271 compass sensor
- S49E Hall effect sensor
- Servo motors (MG90S and MG996R)
- PCA9685 PWM driver
- PCA9548A I2C Multiplexer

- Rechargeable Li-Po battery
- 20A+ Buck converter
- 3D-printed robotic arm structure

Safety Precautions

- Do not touch moving parts during operation.
- Ensure proper battery connection before powering the system.
- Place the robotic arm on a stable surface.
- Avoid operation near water or flammable materials.
- Disconnect power before maintenance.

System Setup Instructions

1. Place the robotic arm on a stable surface.
2. Connect servo motors to the PWM driver.
3. Connect sensors to the I2C mux.
4. Connect the PWM driver and I2C mux to the microcontroller
5. Connect the LiPo battery to the buck converter
6. Set up the buck to give 5V output voltage
7. Connect the buck to the PWM driver
8. Use a separate power source to power the microcontroller
9. Ensure the batteries are charged.
10. Turn on the power supply and wait for system initialization.

Operating Instructions

- Position the sensors on the hand.
- Move the hand slowly and naturally.
- The robotic arm will mimic the hand movements.

Control Examples

- Shoulder movement → Shoulder servo
- Elbow bending → Elbow servo
- Wrist rotation → Wrist servo
- Hand rotation → Base servo
- Finger movement → Gripper operation

Smooth movements improve accuracy.

System Shutdown Procedure

1. Stop all hand movements.
2. Turn off the power supply.
3. Disconnect the battery.
4. Allow the system to cool before storage.

7.3 GitHub Link

https://github.com/intmushfiq/6-DOF-Robotic-Arm

7.4 YouTube Link

https://youtu.be/FEJS1v8x_Gw?si=s2ygygGEdgHiezDW

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Sl. No.	Component Name	Quantity	Total Cost (BDT)	Remarks
1	3D Printed Body Parts	-	4132	Includes base, links, claw
2	MG996R Servo Motor	11	3776	Main joints
3	MG90S Micro Servo	5	510	Smaller joints
4	Servo Driver (8-Channel PWM)	1	400	PWM control
5	MPU6050 (GY-521)	3	610	Motion sensing
6	Magnetometer (GY-271 / GY-511)	2	956	Base rotation
7	Hall Effect Sensor + Magnet	-	540	Gripper control
8	I2C Multiplexer (TCA Module)	1	750	Sensor interfacing
9	Buck Converter	4	1300	Voltage regulation
10	LiPo Battery (3.7V)	1	400	Power supply
11	Jumper Wires & Header Pins	-	280	Connections
12	Bluetooth Module	1	370	Wireless
13	Bearings & Mechanical Parts	-	450	Smooth motion
14	Band, Glue, Cutter	-	670	Assembly

15	Servo Extension Cables	-	240	Wiring
16	Miscellaneous	-	475	Others

8.2 Calculation of Per Unit Cost of Prototype

The total cost required to build the prototype was approximately 17,549 BDT. This includes all components from sensors and motors to mechanical structure and power supply.

Since this was a one-time build, some costs, such as tools, small accessories, and trial components, slightly increased the total expense.

When shared among group members, the cost per person comes out to around ≈ 3,510 BDT, which is reasonable for a project of this scale.

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

If the same system is produced in larger quantities, the cost can be reduced significantly.

Bulk purchasing of components would lower individual prices, especially for motors and sensors. Also, instead of 3D printing, manufacturing techniques like molding can reduce production costs for mechanical parts.

With these optimizations, the estimated cost per unit could be reduced by around 20–30%, making it approximately 12,000 to 14,000 BDT per unit.

Further reduction is possible with better supply chain management and design optimization.

9 Future Work (PO(l))

- Increase degrees of freedom to replicate more accurate hand movement
- Upgrade the gripper into all five fingers replicating model
- Use advanced sensors to improve accuracy and safety
- Use a better stabilization scheme, like a PID controller, for increased stability
- Implementation of wireless communication
- Improvement of mechanical design and materials
- Integration of artificial intelligence
- Human-Machine interface development
- Real-Worl application and deployment

10 References

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